Practice workbook · Explain how villi aid absorption and the effect of damage

Villus and microvillus structure for absorption, and consequences when villi are flattened or nutrients are taken by parasites.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VILLUS-ABSORPTION

1. Question 5 · part c.ii · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c.ii only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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1. Question 5 · part c.ii · 3 marks

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Worked-example retry · Attempt part c.ii only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

(3)

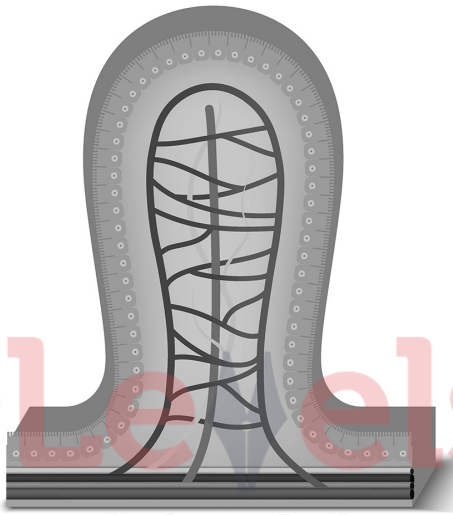
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1. Question 5 · part c.ii · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c.ii only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



(Source: © Dee-sign / Shutterstock)

(i) Name this structure.

(1)

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(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

(3)

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(Total for Question 5 = 12 marks)

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1. Question 5 · part c.ii · 3 marks · Working and answer

Working and answer

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2. Question 4 · part c.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.iii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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P 5 9 9 1 0 A 0 8 2 4

2. Question 4 · part c.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.iii only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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2. Question 4 · part c.iii · 2 marks · Working and answer

Working and answer

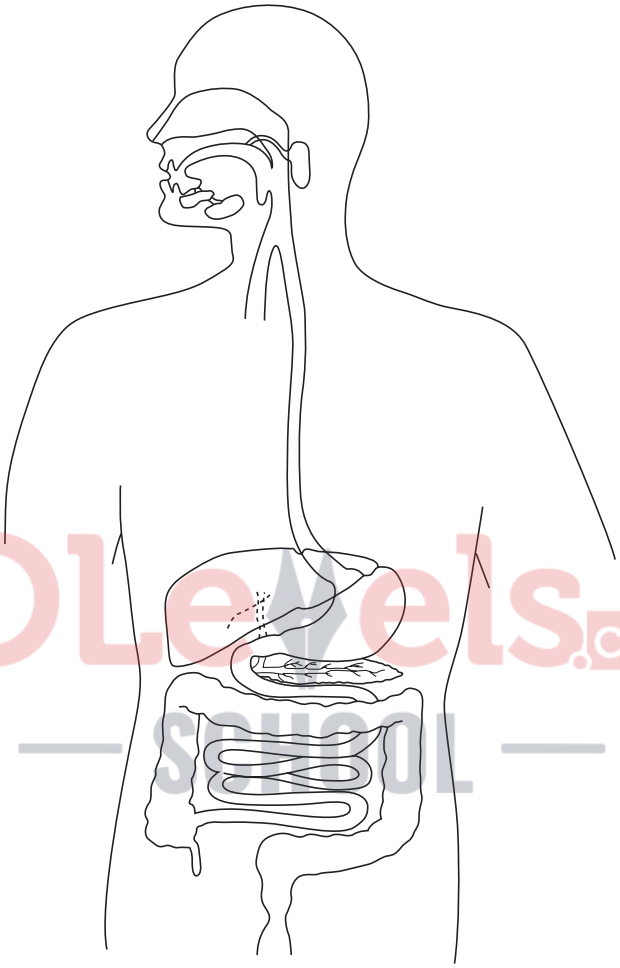
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3. Question 4 · part b · 4 marks

November 2021 Paper 1

Further practice · Attempt part b only.

4 The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

(1)

(ii) Label with a Y the area that has a pH of 2

(1)

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

(1)

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3. Question 4 · part b · 4 marks

November 2021 Paper 1

Further practice · Attempt part b only.

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(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

(4)

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(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

(2)

(Total for Question 4 = 9 marks)

3. Question 4 · part b · 4 marks · Working and answer

Working and answer

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4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

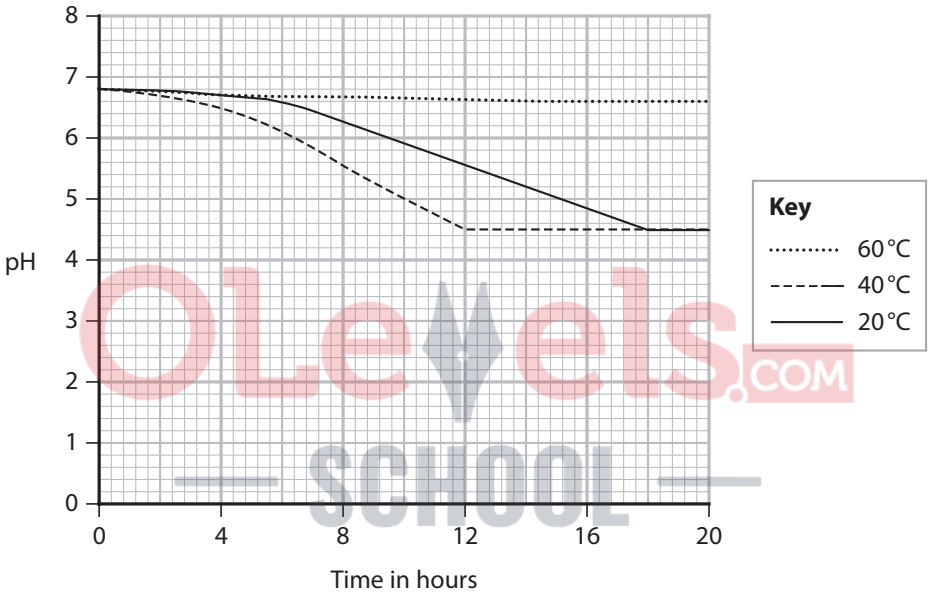
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

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2 .....

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(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

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cooling

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P 7 2 2 7 1 A 0 1 7 2 4

4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

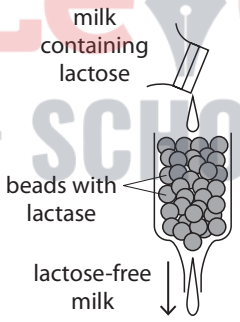
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

(2)

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.



(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

(2)

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4. Question 6 · part c.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(ii) The enzyme lactase breaks down lactose into simpler sugars.

Explain how the structure of the small intestine allows the absorption of these sugars.

(3)

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(Total for Question 6 = 15 marks)

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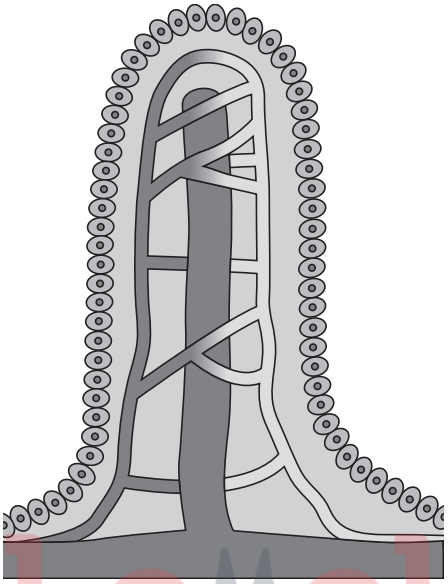
Working and answer

5. Question 7 · part a · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

7 The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

(3)

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5. Question 7 · part a · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

Describe how the immune system of people with coeliac disease damages the lining of the small intestine.

(2)

(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

(3)

(Total for Question 7 = 8 marks)

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5. Question 7 · part a · 3 marks · Working and answer

Working and answer

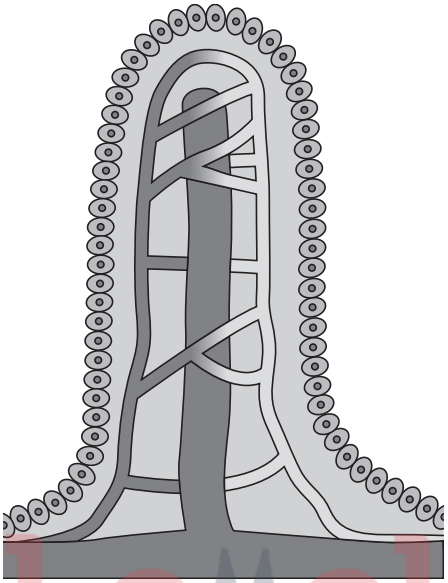
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6. Question 7 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

7 The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

(3)

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6. Question 7 · part b.ii · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

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(2)

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(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

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(Total for Question 7 = 8 marks)

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6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

(3)

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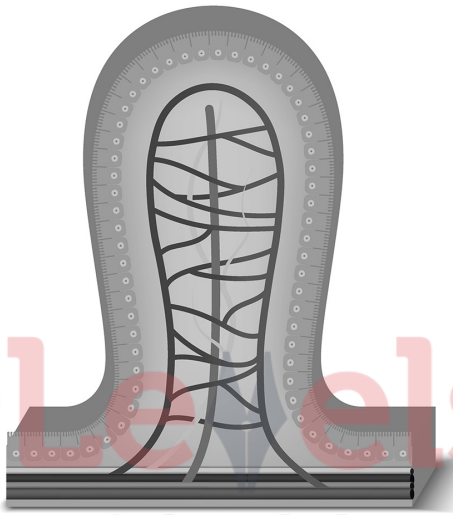
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7. Question 5 · part c.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c.i only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



(Source: © Dee-sign / Shutterstock)

(i) Name this structure.

(1)

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(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

(3)

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(Total for Question 5 = 12 marks)

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7. Question 5 · part c.i · 1 mark · Working and answer

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